

Assignment 1

Applied Generative AI

Part 1: API and Docker

GitHub: <https://github.com/xiangbuchu77/sps-genai-assignment1>

The Module 3 API has been extended with POST `/embedding`. It uses the `en_core_web_lg` model from Module 2 and returns `nlp(word).vector` as a JSON list. The original GET `/` and POST `/generate` routes are also included.

Run with Docker

With Docker running, use these commands:

```
git clone https://github.com/xiangbuchu77/sps-genai-assignment1.git
cd sps-genai-assignment1
docker build -t sps-genai-assignment1 .
docker run --rm -d --name sps-genai-assignment1 \
  -p 127.0.0.1:8000:80 sps-genai-assignment1
```

The first build downloads the spaCy model and dependencies. Once the container is ready, open <http://127.0.0.1:8000/docs> to try the endpoints.

Example request

```
curl -X POST http://127.0.0.1:8000/embedding \
  -H 'Content-Type: application/json' \
  -d '{"word": "apple"}'
```

The response contains word, model, dimensions, and embedding. For apple, it returns 300 values. The first five are approximately -0.36391 , 0.43771 , -0.20447 , -0.22889 , and -0.14227 .

The input must be one alphabetic token. Invalid input returns HTTP 422; a word missing from the model vocabulary returns HTTP 404.

The Docker image was built and tested on linux/arm64. The container started successfully, and the embedding and text-generation endpoints returned valid responses.

To stop the container:

```
docker stop sps-genai-assignment1
```

The code follows the Module 2 word-embedding and word-sampling notebooks and the Module 3 FastAPI activity. Local setup instructions are in the repository README.

Part 2: Rules of Probability

Question 1

Given $P(A) = 0.4$ and $P(B) = 0.3$, with A and B independent:

(a) Independence gives $P(A \cap B) = P(A)P(B) = 0.4 \times 0.3 = \mathbf{0.12}$.

(b) The addition rule gives $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.3 - 0.12 = \mathbf{0.58}$.

Question 2

Given $P(A) = 0.5$, $P(B) = 0.4$, and $P(A | B) = 0.7$:

Independence would require $P(A | B) = P(A)$, since $P(B) > 0$. However, $0.7 \neq 0.5$. Therefore, **A and B are not independent**.

Question 3

Given $P(A) = 0.6$, $P(B | A) = 0.5$, and $P(B | A^c) = 0.2$:

$$P(A^c) = 1 - 0.6 = 0.4.$$

By the law of total probability:

$$P(B) = P(B | A)P(A) + P(B | A^c)P(A^c) = 0.5 \times 0.6 + 0.2 \times 0.4 = 0.30 + 0.08 = 0.38.$$

By Bayes' rule:

$$P(A | B) = P(B | A)P(A) / P(B) = 0.30 / 0.38 = 15/19 \approx \mathbf{0.789474}, \text{ or } \mathbf{78.95\%}.$$

Question 4

Let D denote having the disease and + denote a positive test result. The given rates are $P(D) = 0.02$, $P(+ | D) = 0.95$, and $P(- | D^c) = 0.90$.

Thus, $P(D^c) = 0.98$ and the false-positive rate is $P(+ | D^c) = 1 - 0.90 = 0.10$.

$$P(+) = P(+ | D)P(D) + P(+ | D^c)P(D^c) = 0.95 \times 0.02 + 0.10 \times 0.98 = 0.019 + 0.098 = 0.117.$$

$$P(D | +) = P(+ | D)P(D) / P(+) = 0.019 / 0.117 = 19/117 \approx \mathbf{0.162393}, \text{ or } \mathbf{16.24\%}.$$

The probability is about 16.24%. Although the test detects most cases, the disease is rare, so false positives account for most positive results.

Question 5

x	P(X = x)	xP(X = x)	x ² P(X = x)
85	0.375	31.875	2709.375
90	0.375	33.750	3037.500
95	0.125	11.875	1128.125
100	0.125	12.500	1250.000
Total	1.000	90.000	8125.000

(a) $E[X] = \sum xP(X = x) = 31.875 + 33.750 + 11.875 + 12.500 = \mathbf{90}$.

(b) $E[X^2] = \sum x^2P(X = x) = 8125$.

$\text{Var}(X) = E[X^2] - (E[X])^2 = 8125 - 90^2 = 8125 - 8100 = \mathbf{25}$ (score points squared).

(c) For the sample {85, 90, 85, 95, 90, 85, 100, 90}:

Sample mean = $(85 + 90 + 85 + 95 + 90 + 85 + 100 + 90) / 8 = 720 / 8 = \mathbf{90}$.

The sample mean equals $E[X] = 90$. In this sample, the relative frequencies match the given probabilities: 3/8, 3/8, 1/8, and 1/8.

Question 6

(a) With probabilities (0.4, 0.3, 0.2, 0.1), entropy in bits is:

$$H(X) = -\sum p(x)\log_2 p(x)$$

$$= -[0.4 \log_2(0.4) + 0.3 \log_2(0.3) + 0.2 \log_2(0.2) + 0.1 \log_2(0.1)]$$

$$= 0.528771 + 0.521090 + 0.464386 + 0.332193$$

$$\approx \mathbf{1.846439 \text{ bits}}$$

(b) If the four messages are equally likely, each has probability 1/4:

$$H(X) = -4 \times (1/4)\log_2(1/4) = -\log_2(1/4) = \mathbf{2 \text{ bits}}$$

Equal probabilities give the maximum entropy for four messages.